

Suppose the number of accidents at a location in a week is a Poisson random variable with parameter $\lambda = 2$ and the amount of damage in these accidents are independent and identically distributed with mean 2,000 and variance 4,000,000. Suppose the amount of damage in each accident is independent of the number of accidents in a week.

1. Find the expected value of the total amount of damages in all accidents in a week.
2. Find the variance of the total amount of damages in all accidents in a week.